

APOLLO SOLAR VENTURES INC.

NUTECH Hardware & Software Solutions

NUTECH School System

Unified Cashless POS & Two-Factor Attendance

Proof-of-Concept — Full Project Report

Prepared for
Bernard P. Jocson — Founder & President

Status: **Integration & Testing Phase**

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Address: 1864 District 4A, Brgy. Marauoy, Lipa City, Batangas 4217

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1. Executive Summary

The system is a single, unified school platform that runs two functions on **one shared backend**: a **cashless cafeteria POS** and a **two-factor (RFID + face) access-control attendance gate**. One person equals one **ACCOUNT** — one RFID card, one wallet, and one face record. The card UID is the primary key across every surface; the face is a secondary check at the gate.

Where the project stands:

- **Front-end apps:** built and working. Three HTML/PWA surfaces, all validated.
- **RFID hardware:** working on all three apps (POS confirmed; gate and portal fixed with auto-reconnect).
- **Fingerprint:** removed for now (DigitalPersona device offline). Card-only.
- **Airtable base:** built and seeded with test data; schema complete.
- **n8n workflow:** core POS branches built and working; access-control and dashboard branches in progress.
- **Outstanding:** finish remaining n8n branches, delete one Airtable column, end-to-end test, then flip apps from demo to live.

2. System Architecture

Three capture surfaces send HTTPS POST requests to a single n8n workflow, which routes each request by its **action** value to the correct branch and reads or writes the Airtable base.

Capture surfaces	Orchestration	Data
portal.html (PWA)	n8n workflow	Airtable base
gate.html (gate station)	"Route by action" Switch	"Nutech POS"
cafeteria-pos.html (cashier)	single webhook path	apprpYxg7leO7JXKJ

- **Webhook:** <https://bernard100.app.n8n.cloud/webhook/cafeteria>
- **Identity hub:** the ACCOUNT table. Role tables (STUDENT / PARENT / PERSONNEL) branch off it.

3. The Three Applications

3.1 cafeteria-pos.html — staffed cashier terminal

- **Purpose:** canteen checkout. Tap card, see balance, ring up items, pay, receipt; plus top-up and card enrollment.
- **Hardware:** Arduino + PN532 RFID over USB (Web Serial). Confirmed working.
- **State:** live. Fingerprint checkout removed — card-only.
- **Actions:** verify, purchase, topup, enroll, menu.

3.2 gate.html — entrance station (two-factor attendance)

- **Purpose:** RFID tap, identify person, 1:1 face match (browser camera), then write attendance (Time In/Out, Present/Tardy).
- **Hardware:** RFID via USB HID or Arduino over Web Serial (auto-reconnect); camera via getUserMedia.
- **State:** demo toggle in settings; Test-connection button.
- **Actions:** gate-verify, gate-punch.

3.3 portal.html — phone / PWA for everyone

- **Purpose:** role-based app — Admin (dashboard, register, ordering, credit approvals, attendance, people, reports, settings), Parent (credit requests with proof), Student, Personnel (salary-deduction view).
- **Hardware:** RFID reader on the Register screen (USB HID or Arduino + Web Serial, auto-reconnect, auto-arm).
- **State:** Settings page with demo/live switch, webhook field, Test connection, POS-style reader selector. Fingerprint capture removed.
- **Actions:** register-person, credit-request, credit-requests, credit-approve, admin-overview, client-data, salary, menu, account-info.

All three share branded styling, a **demo mode** that runs fully offline on sample data, and a **live mode** that talks to n8n. Demo versus live is a toggle chosen ahead of time, not an automatic fallback.

4. Hardware Status

Device	Where	Method	Status
Arduino + PN532 RFID	POS PC	Web Serial (USB, 115200)	Confirmed
Arduino + PN532 RFID	Gate PC	Web Serial + auto-reconnect	Fixed
Arduino + PN532 RFID	Portal PC	Web Serial + auto-reconnect	Fixed
USB HID RFID reader	Gate / Portal	Keyboard-style UID	Supported
Camera (face match)	Gate	getUserMedia + face-api.js	Works
DigitalPersona 4500	—	local bridge (9001)	Removed
Magnetic gate relay	Gate	local relay bridge	Theoretical
Receipt printer XP-58IIH	POS	print-bridge (9002)	Optional

Critical constraint — one reader per page. A USB reader is owned by a single browser page at a time. On one laptop, only one app can hold the Arduino at once (close the POS tab before using the gate). In production each station has its own reader, so this is moot.

Arduino sketches print one UID per line at 115200 baud — the contract all three apps expect.

5. Airtable Base — “Nutech POS”

Base ID: apprpYxg7leO7JXKJ

Tables

Table	Role
ACCOUNT	Identity hub — AccountID, RFID UID, FACE_BIO_ID, Balance, Status, Owner Name/Type, role links.
STUDENT	StudentID, Name, Section, Contact, Email, IsEnrolled, Address, Sex.
PARENT	ParentID, Name, Contact, Email, Address.
PERSONNEL	Personnel_ID, Name, Contact, Email, Role, Salary.
MENU	Item, Price, Category, Available, Stock On Hand, Reorder Level, Stock Status.
TRANSACTIONS	TxnRef, Type, AccountID, Amount, Balance After, Items, By, Created, Sale Date.
SALE_ITEMS	One row per cart line — Item, Qty, Unit Price, Line Total, Cashier, Account ID.
DTR	Daily time record — Date, Time In/Out, Break, Status, Account ID.
ATTENDANCE	Raw taps — Direction, Method (incl. Card+Face), Device, Timestamp.
CREDIT_REQUESTS	Parent top-up requests with photo proof, Status, Decided By.
ACTIVITY_LOG	Audit trail — Actor, Action, Target Ref, Details, Amount, Device.

Seeded test fixtures (delete when done)

RFID	AccountID	Person	Status
0A1B2C3D	ACC-TEST-MARIA	Maria Santos (P320)	Active
1B2C3D4E	ACC-TEST-RDC	R. dela Cruz (Teacher)	Active

RFID	AccountID	Person	Status
2C3D4E5F	ACC-TEST-TROY	Troy Sato	Lost Card

6. n8n Workflow

Every branch is built from four node types: **HTTP Request** (Airtable REST API), **Code** (reshape), **IF** (branch), and **Respond**. Auth uses the Airtable Token API credential, carried automatically when nodes are copied.

6.1 Built and working

Action	Function
verify	POS card tap — identity and balance.
purchase	Ring up sale; deduct balance; Split Cart → SALE_ITEMS → stock decrement → Log Activity.
topup	Add to wallet; log transaction.
enroll	Link a blank card to an account.
menu	Return menu items.
account-info	Return wallet / identity (this is “check balance”).
reassign	Move a card to a new owner.

6.2 In progress / to build

Action	Function	Note
gate-verify	Gate tap — identity, face on file, today's punches.	build first
gate-punch	After face match — write DTR + ATTENDANCE.	IF: update/create
register-person	Create ACCOUNT and role row together.	replaces register
credit-request	Parent request + photo upload.	content.airtable.com
credit-requests	List pending requests.	
credit-approve	Approve — credit wallet + log.	IF on decision
admin-overview	Dashboard totals.	JSON fix (\$7)
client-data	One person's history.	
salary	Personnel monthly cafeteria spend.	
nightly absent	Mark Absent.	scheduled

Note on “attendance”: the old single-tap attendance branch is obsolete — replaced by gate-verify + gate-punch. The word “Attendance” elsewhere is only a category label written to the ACTIVITY_LOG Action field, not a Switch branch.

7. Known Issue Under Test

“Invalid JSON in Response Body.” In a Respond-to-Webhook node set to Respond With = JSON, handing it a pre-stringified value trips n8n's JSON validation.

Fix: set Respond With → **First Incoming Item** on every Respond node. This removes the Response Body field entirely; n8n serializes the upstream object itself. The Code nodes are already correct.

Companion symptom “Cannot reach n8n” means the workflow is not Active yet — the production URL is dead until the top-right toggle is on. Both clear once the workflow is Active and the Respond nodes use First Incoming Item.

8. Outstanding Tasks

For Bernard

1. **Delete the FINGERPRINT_ID column** in ACCOUNT (Airtable's API can't delete fields — a 2-click manual step: column header ► Delete field). The field is empty; no data lost.
2. **Finish the n8n branches** in §6.2, in the stated build order.
3. **Set every Respond node to First Incoming Item** (§7).
4. **Activate the workflow** and curl-test each branch against the test fixtures.
5. **Flip apps to live** — portal Settings → live; gate settings → uncheck demo (POS already live).
6. **Optional:** delete the unused bio-enroll / bio-template branches.

Already done

- All schema built and seeded; RFID working on all three apps with auto-reconnect; fingerprint removed from all apps; demo/live toggles on every app; POS-style reader UI in portal; full click-by-click n8n build guides written.

9. Safe Testing Approach

- Use the **test webhook URL** (/webhook-test/cafeteria) with “Listen for test event” while building — fires only while you watch.
- Test **read-only branches first** (gate-verify, menu, account-info, dashboards) — they can't harm data. Then row-creating branches. Then balance-changing branches (topup, purchase, credit-approve), only against Maria, noting her ₱320 start.
- Eyeball the Airtable row after each write. Airtable revision history is a safety net.
- Keep apps in **demo** until curl proves each branch.

10. Go-Live Checklist

- ☐ FINGERPRINT_ID column deleted in Airtable
 - ☐ All §6.2 branches built
 - ☐ Every Respond node set to First Incoming Item
 - ☐ Workflow toggled Active (only one workflow owns /cafeteria)
 - ☐ Each branch curl-tested against fixtures
 - ☐ Apps flipped to live (portal + gate), Test connection passes
 - ☐ Test fixtures deleted, real students and menu loaded
 - ☐ Each station has its own RFID reader
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